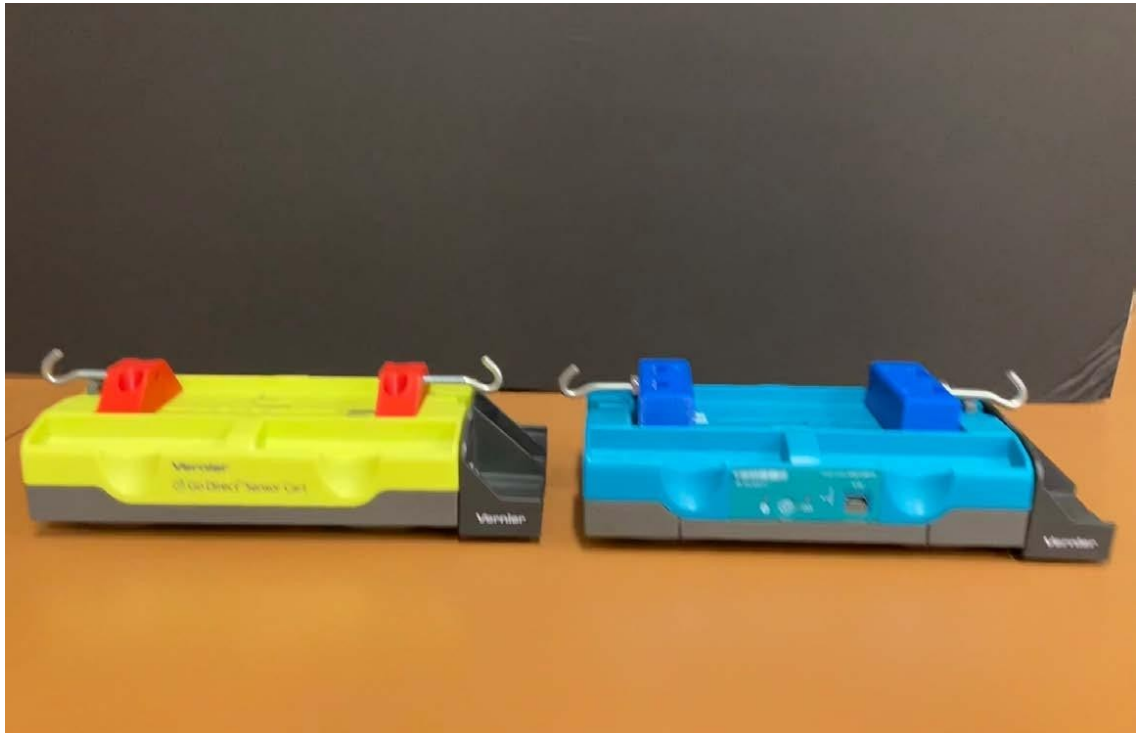
Free vibration of a spring–mass system

From the real cart to a testable mathematical model

What makes it oscillate?
Why does it stop?
What can one position record tell us?



We will derive every analysis formula from the same equation of motion.

0 – 1 min

Why engineers measure a free response



NASA / GSFC: TESS vibration testing

Spacecraft and equipment testing

Measured frequency checks stiffness and moving mass.

Decay rate reveals energy losses and settling behavior.

The same ideas guide suspension tuning and machinery diagnostics.

Lab 4 isolates these ideas in a single horizontal cart.

One cart, one model, four things to measure

Observe

The encoder gives a time series of cart position, X(t).

→

Explain

Newton’s law gives mass, stiffness, and damping parameters.

→

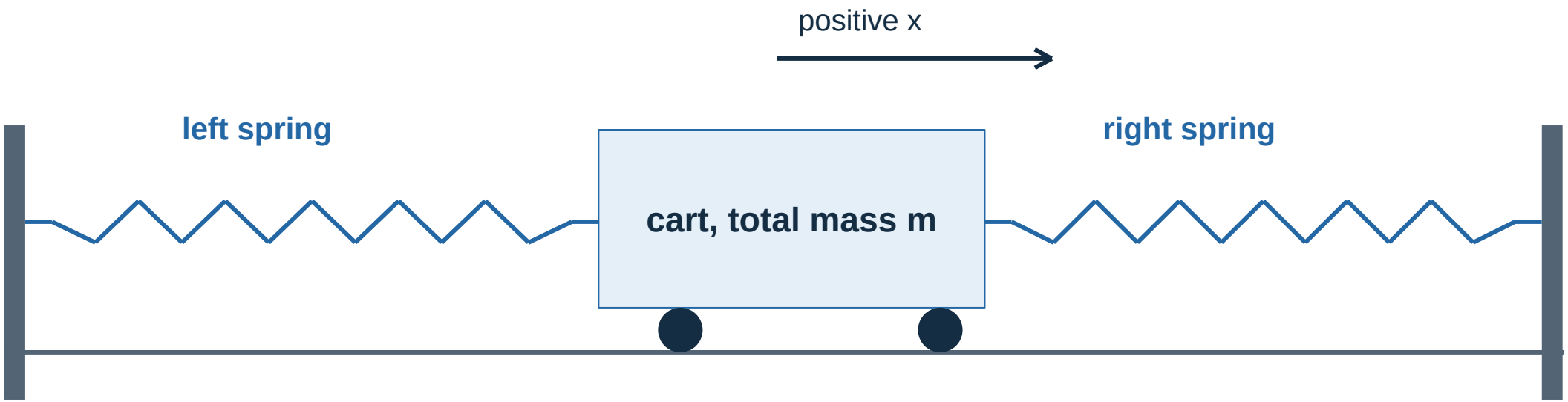
Estimate

Peak spacing, peak decay, and FFT check different features.

Every measured formula will be traced back to the equation of motion.

2–3 min

The physical test: a cart between two springs



Both springs must remain engaged near equilibrium; verify preload.

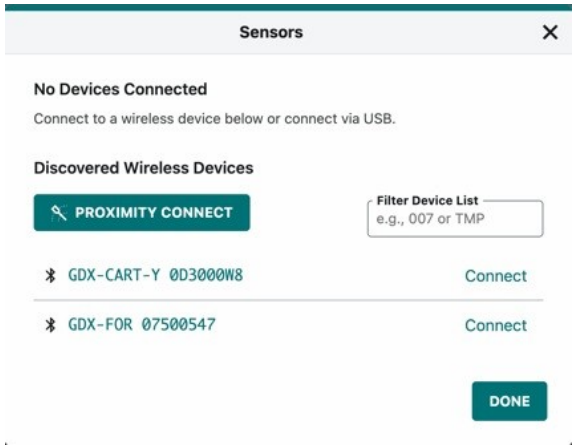
Baseline: remove added magnetic brakes. Residual rolling and air losses remain.

How position reaches the computer



Encoder wheel

Wheel rotation becomes position x(t).
The encoder wheel must roll without slip.



Vernier Graphical Analysis

Select the cart by its serial number.
Record time and position, then export CSV.

The sensor supplies x(t). The analysis code estimates frequency and damping.

Spring hardware and brake hardware have different roles





Coil spring

Stores elastic energy and provides restoring force.
Illustrative Vernier set. Calibrate the actual lab springs.

Eddy-current brake

Holds removable magnets near the track.
The lab varies magnet count to change damping.

Measure the mass of attached hardware. A magnet count is different from a brake-assembly count.

The experiment changes one property at a time

Base	original mass, original spring pair, no added brakes
Two mass cases	same springs, add one mass and then a second
Two damping cases	same springs, add two and then four brake magnets
Two stiffness cases	return to base mass and change spring combinations

We derive the predictions first. Prior-class measurements follow the analysis methods.

The coordinate and all model variables

$$x(t) = X(t) - X_{\text{eq}}$$

$$\dot{x} = \frac{dx}{dt}, \qquad \ddot{x} = \frac{d^2x}{dt^2}$$

x: displacement from equilibrium [m]

x-dot: velocity [m/s]

m: total moving mass [kg]

x-double-dot: acceleration [m/s²]

Left / right spring stiffnesses [N/m]

One degree of freedom: one coordinate describes the cart.

X: encoder position. Xeq: equilibrium position [m]. t: time [s].

A level track balances weight with the normal force. We model horizontal motion.

Why the spring stiffnesses add

For a small positive displacement x:

Left spring stretches more

$$\Delta F_L = -k_L x$$

Right spring stretches less

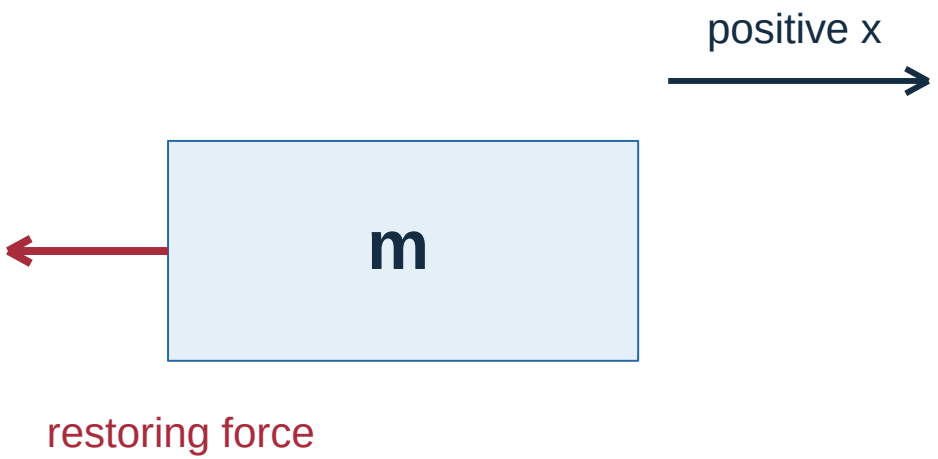
$$\Delta F_R = -k_R x$$

$$F_s = -(k_L + k_R)x = -k_{\text{eq}} x$$

$$k_{\text{eq}} = k_L + k_R$$

The constant preload forces cancel at equilibrium. Their changes provide the restoring force.

Newton’s second law gives the undamped equation



$$\sum F_x = m\ddot{x}$$

$$-k_{\text{eq}}x = m\ddot{x}$$

$$m\ddot{x} + k_{\text{eq}}x = 0$$

After release the external forcing is zero. The initial displacement starts the motion.

How the equation produces a sinusoid

Try a sinusoidal motion with unknown angular frequency ω:

$$x = A\cos(\omega t + \phi)$$

$$\ddot{x} = -\omega^2 A\cos(\omega t + \phi)$$

$$\left(k_{\text{eq}} - m\omega^2\right)A\cos(\omega t + \phi) = 0$$

$$\omega_n = \sqrt{\frac{k_{\text{eq}}}{m}}$$

A is displacement amplitude [m]. φ is initial phase [rad]. Initial conditions set both.

Angular frequency, frequency, and period

$$\omega_n = \sqrt{\frac{k_{\text{eq}}}{m}}, \qquad f_n = \frac{\omega_n}{2\pi}, \qquad T_n = \frac{1}{f_n}$$

Angular natural frequency [rad/s]: phase advance per second

Natural frequency [Hz]: complete cycles per second

Natural period [s]: time between successive equal-sign peaks

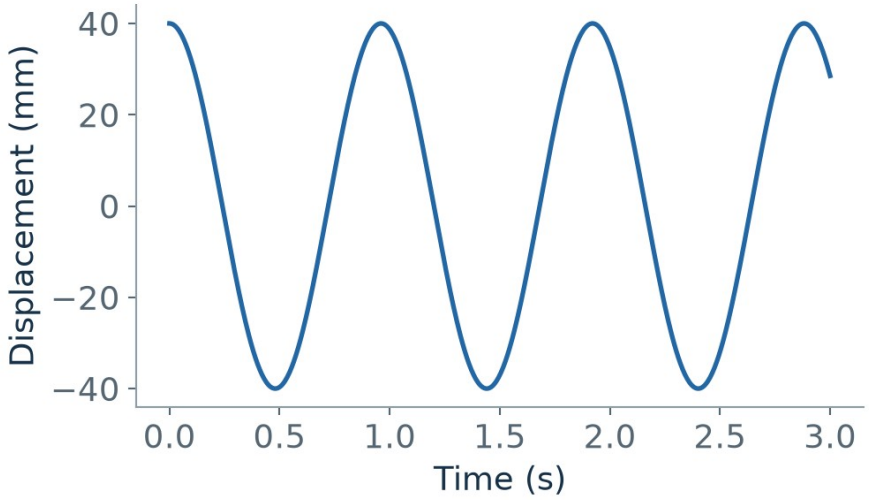
$$m = 0.50 \text{ kg}, \;\; k_{\text{eq}} = 21.4 \text{ N/m} \quad \Rightarrow \quad f_n = 1.04 \text{ Hz}$$

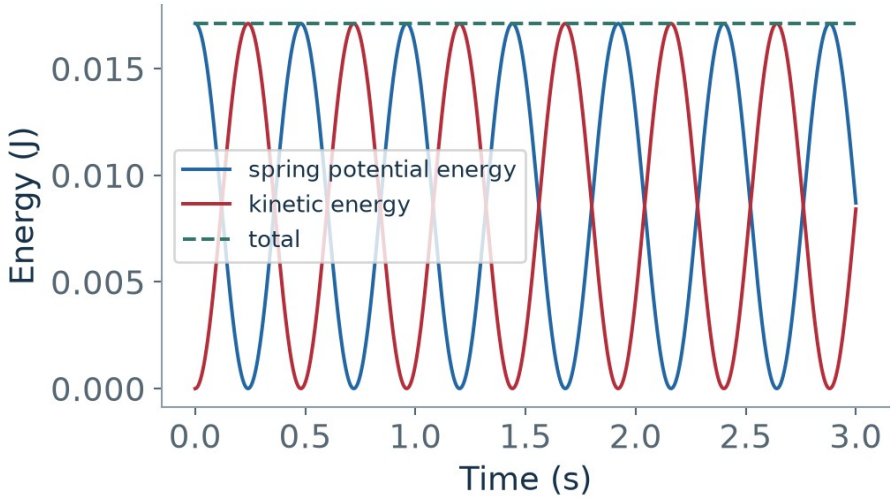
Example values illustrate the model. Use the measured moving mass and calibrated springs.

Energy moves between the springs and the cart

$$E = \tfrac{1}{2} m \dot{x}^2 + \tfrac{1}{2} k_{\text{eq}} x^2$$

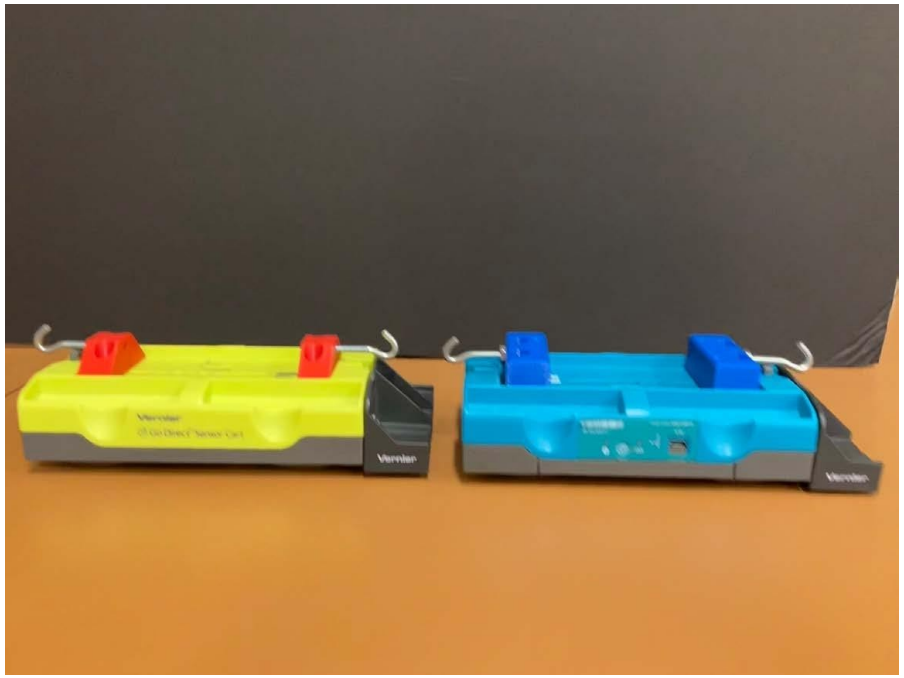
Spring energy is the area under the linear force–extension curve. E is total energy [J].





At a turning point: spring energy is largest. At equilibrium: kinetic energy is largest.

The real damper is a magnetic brake



Black end attachment

Magnetic brake hardware on the real cart.
Confirm magnet count with the GTA.

Magnets move over the conducting track.

Induced eddy currents oppose the motion.

Mechanical energy becomes heat.

$$F_d \approx -c\dot{x}$$

c is an equivalent damping coefficient [N·s/m]. A dashpot symbol represents this force law.

Damping force and energy loss come from the same law

$$F_d = -c\dot{x}$$

Opposes the instantaneous velocity.

$$P_d = F_d\dot{x} = -c\dot{x}^2$$

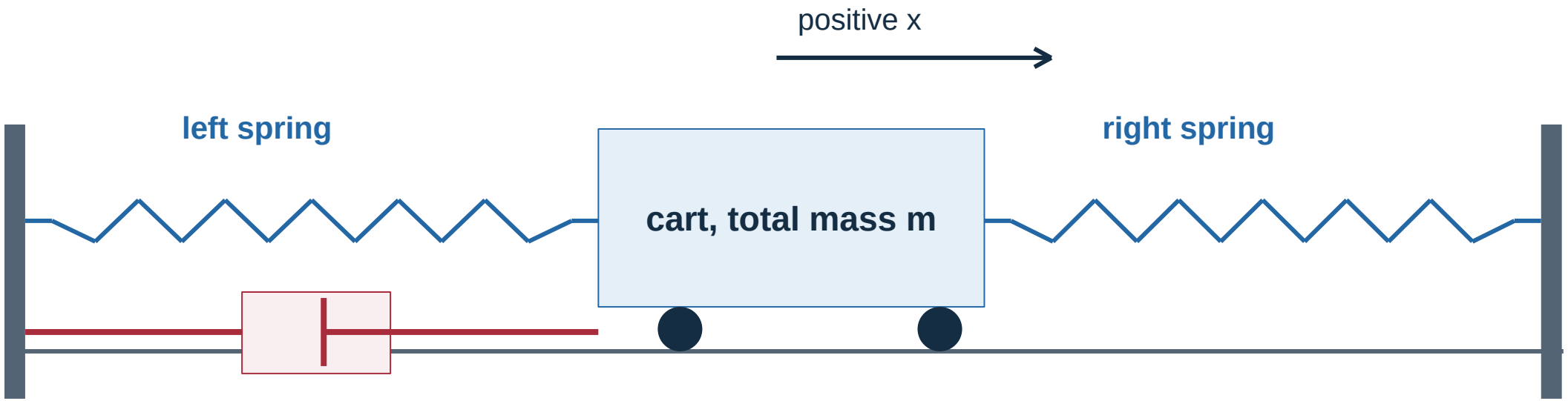
Signed damping power [W]. Negative power removes mechanical energy.

$$\frac{dE}{dt} = -c\dot{x}^2 \leq 0$$

Loss is largest where speed is largest.
It is zero instantaneously at a turning point.

A smaller amplitude means less mechanical energy remains in the oscillation.

Adding the brake changes the force balance



$$-k_{\mathrm{eq}}x - c\dot{x} = m\ddot{x}$$

Adding a velocity-dependent force gives the damped equation. Set c = 0 to recover the undamped model.

Standard form is only a change of notation

1. Divide Newton’s equation by m.

$$\ddot{x} + \frac{c}{m}\dot{x} + \frac{k_{eq}}{m}x = 0$$

2. Define the two coefficients.

$$\omega_n^2 = \frac{k_{eq}}{m}, \qquad 2\zeta\omega_n = \frac{c}{m}$$

3. Substitute those definitions.

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2 x = 0$$

$$\zeta = \frac{c}{2m\omega_n} = \frac{c}{2\sqrt{mk_{eq}}}$$

ζ is the dimensionless damping ratio. This is the same physical equation, rewritten.

The complete chain: forces → ODE → response

Forces [N]

$$F_s = -(k_1 + k_2)x, \quad F_d \approx -c\dot{x}$$

Equation [N]

$$m\ddot{x} + c\dot{x} + (k_1 + k_2)x = 0$$

Parameters

$$\omega_n = \sqrt{(k_e q/m)}, \quad \alpha = c/(2m)$$

Observable

$$x(t) = Ae^{(-\alpha t)} \cos(\omega_d t - \varphi)$$

$k_e q = k_1 + k_2$; every term in Newton's equation has units of force.

20–21 min

The characteristic equation tells us the response shape

Try an exponential motion. The unknown root s has units of inverse seconds [1/s].

$$x = e^{st}, \qquad \dot{x} = se^{st}, \qquad \ddot{x} = s^2 e^{st}$$

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

$$s = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

The root’s real part controls decay. Its imaginary part controls oscillation.

The decaying sinusoid is the solution of those roots

$$0 < \zeta < 1: \quad s = -\zeta\omega_n \pm i\omega_d, \qquad \omega_d = \omega_n\sqrt{1-\zeta^2}$$

$$x(t) = e^{-\zeta\omega_n t}[C\cos(\omega_d t) + D\sin(\omega_d t)]$$

Equivalently, combine C and D into an amplitude A and phase φ:

$$x(t) = Ae^{-\zeta\omega_n t}\cos(\omega_d t - \phi)$$

Damped angular frequency: radians per second. C, D and A have units of length. $i^2 = -1$.

Does damping change the natural frequency?

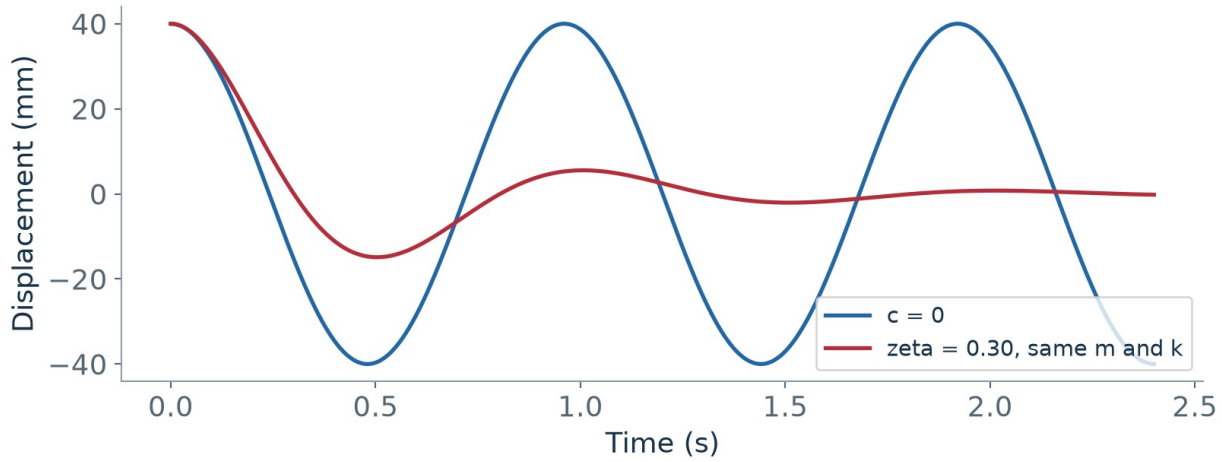
$$\omega_n = \sqrt{\frac{k_{eq}}{m}}, \qquad \omega_d = \omega_n\sqrt{1-\zeta^2}$$

For fixed mass and stiffness:

The undamped natural frequency stays fixed.

The damped frequency falls as damping increases.

At ζ = 0.10, the frequency drop is only 0.5%.



Brake magnets add mass too. Measure that mass before attributing every frequency shift to damping.

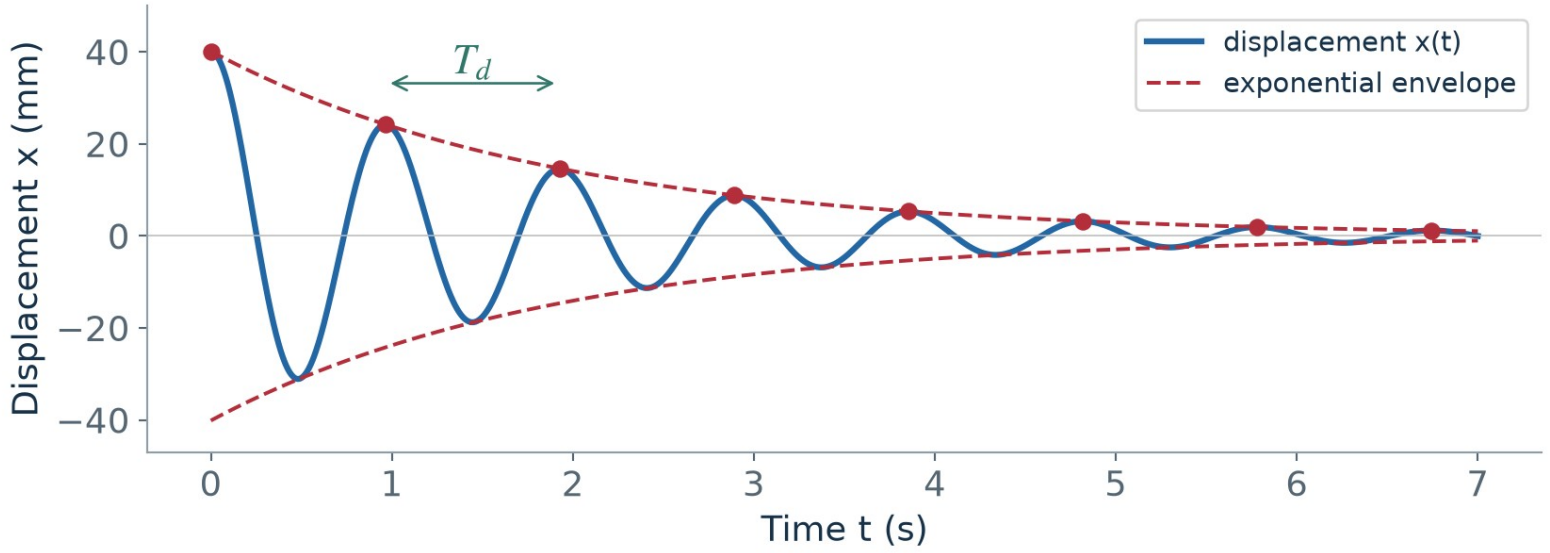
Four frequency/decay quantities are not interchangeable

ω_n [rad/s]	Undamped natural angular frequency: $\sqrt{(k_e q/m)}$.
ω_d [rad/s]	Oscillation angular frequency: $\sqrt{(\omega_n^2 - \alpha^2)}$, for $0 < \zeta < 1$.
α [1/s]	Exponential decay rate: $c/(2m)$, not an oscillation frequency.
fFFT [Hz]	Location of a discrete spectrum peak: an approximate check of $f_d = \omega_d/(2\pi)$.

An FFT peak does not independently determine damping or stiffness.

25–26 min

The envelope shows the size of the remaining oscillation



$$\alpha = \zeta\omega_n = \frac{c}{2m}$$

$$|x(t)| \le Ae^{-\alpha t}$$

α [1/s] sets the decay rate. Equal-sign peaks have the same exponential decay factor.

Blue: displacement. Red dashed curves: exponential bounds. Red dots: selected positive peaks (not necessarily on the envelope).

Peak spacing gives the damped frequency

$$T_d = \frac{t_{i+N} - t_i}{N} \qquad f_d = \frac{1}{T_d}, \qquad \omega_d = 2\pi f_d$$

Use the recorded times of two equal-sign peaks, in seconds.

N counts complete cycles between the selected peaks.

Example: six peaks contain five complete cycles.

Average over several cycles to reduce cursor and sampling error.

Peak spacing measures the damped frequency. Recovering the undamped frequency requires damping.

Peak-height ratios isolate the decay rate

$$p_i = Be^{-\alpha t_i}, \qquad p_{i+N} = Be^{-\alpha(t_i + NT_d)}$$

$$\frac{p_i}{p_{i+N}} = e^{\alpha N T_d}$$

Take the natural logarithm and divide by N:

$$\delta = \frac{1}{N}\ln\left(\frac{p_i}{p_{i+N}}\right) = \alpha T_d$$

p is the magnitude of a selected peak [m]. B is its constant scale [m]. δ is the dimensionless decay per cycle.

The damping ratio follows from δ and the period

$$\delta = \alpha T_d = (\zeta \omega_n)\frac{2\pi}{\omega_d}$$

$$\delta = \frac{2\pi\zeta}{\sqrt{1-\zeta^2}}$$

$$\delta^2(1-\zeta^2) = 4\pi^2\zeta^2$$

$$\zeta = \frac{\delta}{\sqrt{4\pi^2+\delta^2}}$$

The final expression is an algebraic rearrangement of the envelope decay relation.

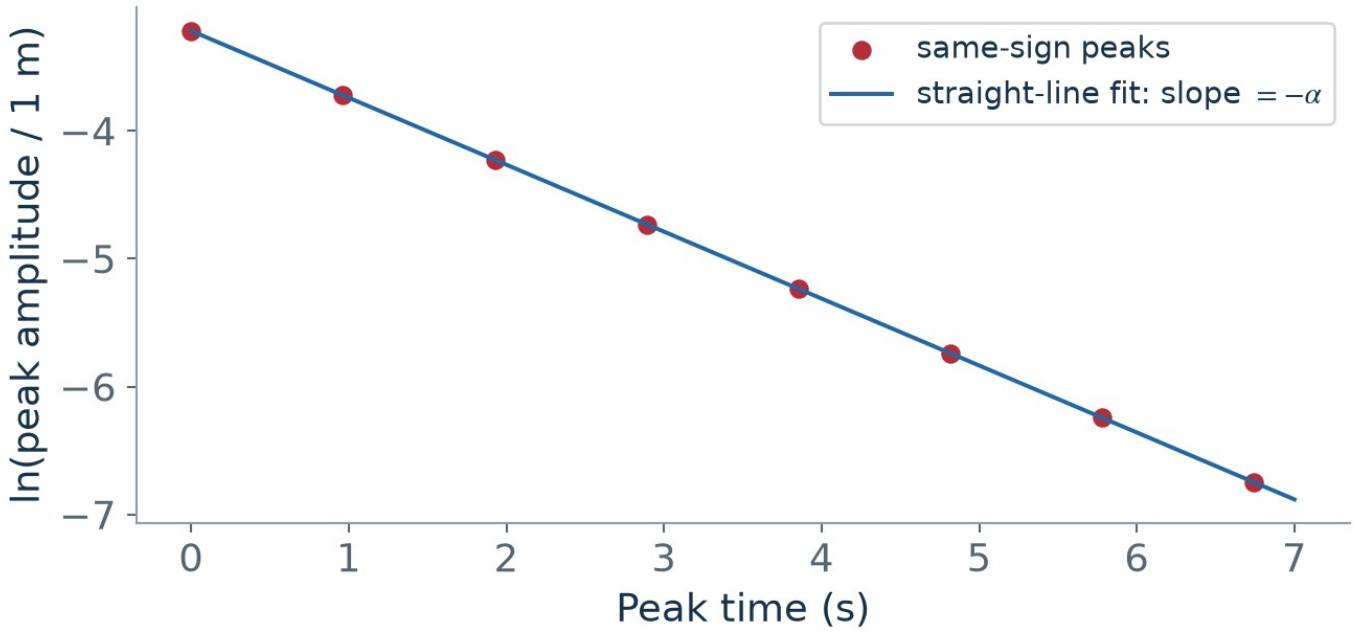
One measured record → parameter estimates

Peak times	Six same-sign peaks span 5.00 s → T_d = 1.00 s.
Peak decay	0.040 m → 0.020 m over five cycles → $\delta = \ln(2)/5 = 0.1386$.
Decay rate	$\alpha = \delta/T_d = 0.1386\ s^{-1}$; $\omega_d = 2\pi/T_d = 6.283$ rad/s.
Model values	$\omega_n = \sqrt{(\omega_d^2 + \alpha^2)} = 6.285$ rad/s; $c = 2m\alpha = 0.139$ N·s/m for m = 0.50 kg.

Illustrative calculation only — not an experimental measurement.

32–33 min

A log-peak fit uses more of the measured record



$$\ln(p/p_{\mathrm{ref}}) = b - \alpha t$$

Fit a straight line through several reliable peaks.

Slope = −α [1/s]
Intercept b sets the starting level.

Reference amplitude: 1 m, so the logarithm has a dimensionless argument.

Exclude late peaks near the noise floor. Curvature signals drift or a poor damping model.

From measured decay and period to physical parameters

From peak timing

$$\omega_d = \frac{2\pi}{T_d}$$

From the log-peak slope

$$\alpha = -\text{slope}$$

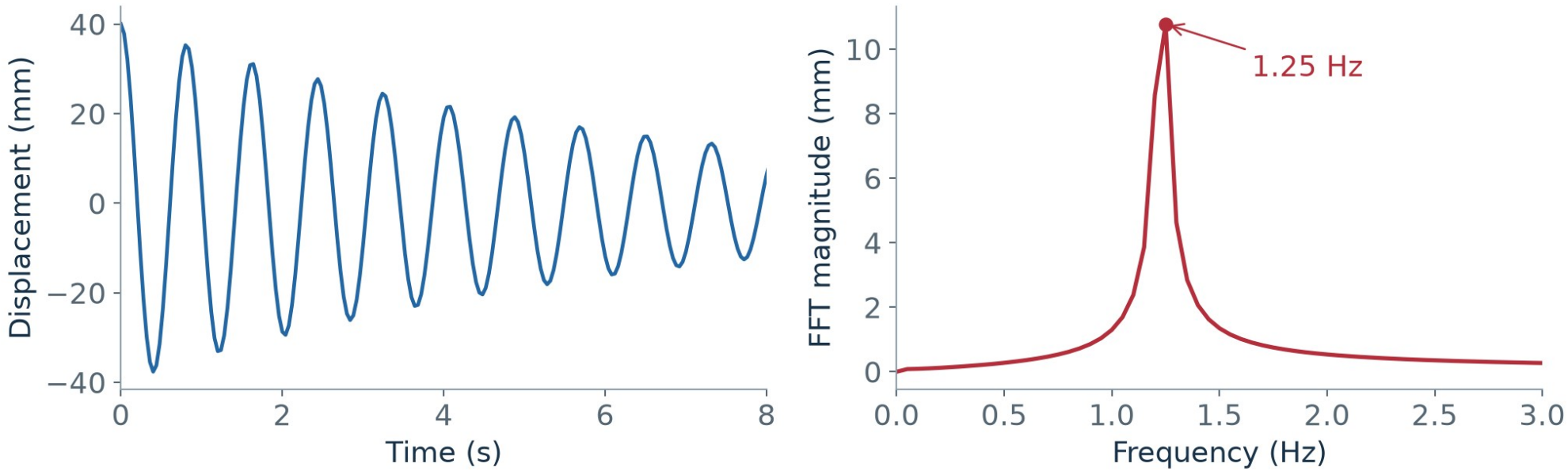
Combine the two measurements

$$\omega_n = \sqrt{\omega_d^2 + \alpha^2}$$

$$\zeta = \frac{\alpha}{\omega_n}, \qquad c = 2m\alpha$$

Measured period and decay rate now give the undamped frequency, damping ratio, and damping coefficient.

The FFT checks which frequency dominates the motion



Blue: first 8 s of a 20 s model record.

Red: its computed FFT magnitude.

The FFT compares the record with trial sinusoids. A matching repetition gives the strongest coefficient.

The dominant peak lies near the damped frequency. Finite duration, decay, and FFT-bin spacing affect the exact peak.

Spring calibration is a separate vertical experiment

Hang a known moving mass from one spring. Use g = 9.81 m/s².

$$k\Delta_0 = m_h g$$

$$m_h\ddot{q} = m_h g - k(\Delta_0 + q) = -kq$$

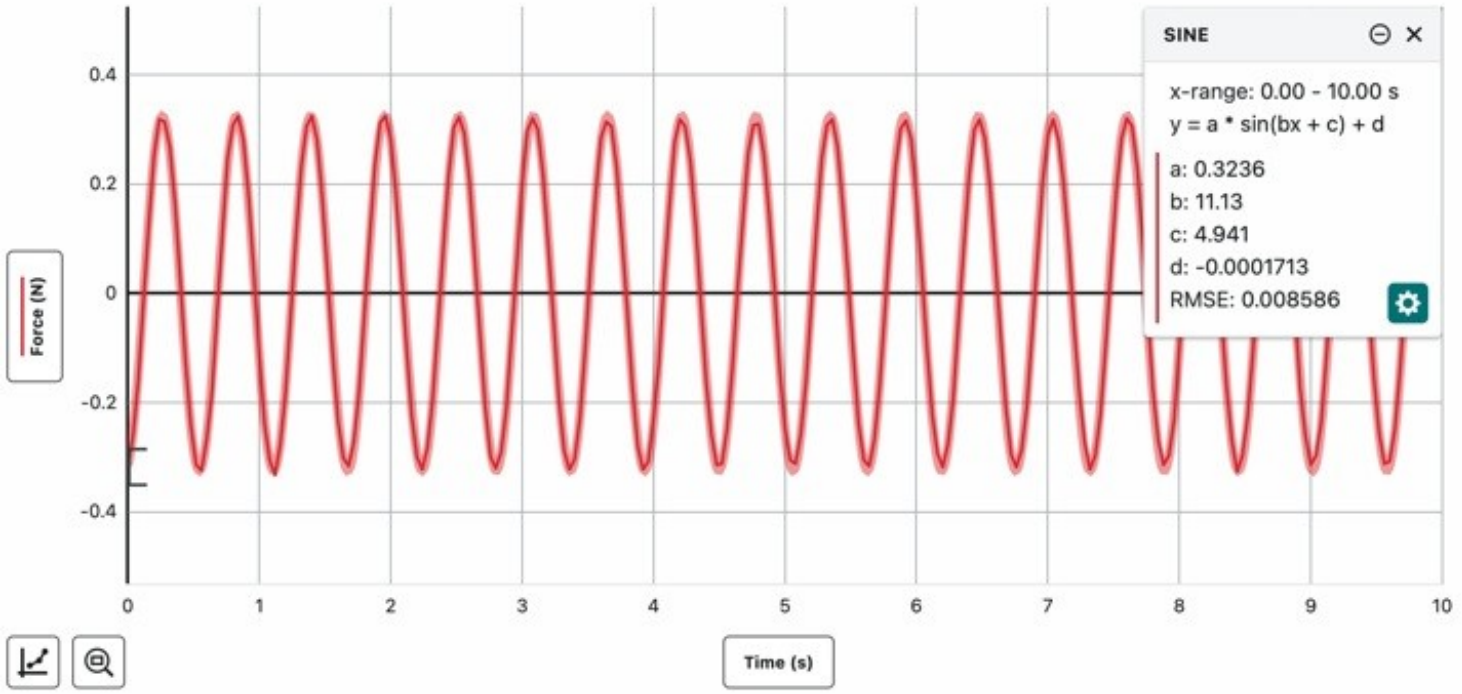
$$m_h\ddot{q} + kq = 0$$

$$\omega_{n,\,s} = \sqrt{\frac{k}{m_h}}$$

q [m]: vertical displacement from the hanging equilibrium.
Static extension [m]: Δ_0. Include the hanger and all moving attachments in the hanging mass.

Small damping makes the measured frequency close to the undamped frequency. Spring mass must be small.

What the sinusoidal fit in Graphical Analysis reports



$$F(t) = a\sin(bt + c_{\text{fit}}) + d$$

a: force amplitude [N]
b: angular frequency [rad/s]
phase parameter: c in the fit window [rad]
d: force offset [N]

$$k \approx m_h b^2$$

Use b directly. Do not multiply b by 2π again.

The screenshot uses c for phase. It is unrelated to the damping coefficient c in the motion model.

The model now predicts the seven-case trends

Increase moving mass	**natural frequency decreases** The decay rate also decreases if the damping coefficient is fixed
Increase stiffness	**natural frequency increases** The decay rate stays fixed if mass and damping coefficient stay fixed
Increase damping	**decay becomes faster** At fixed mass and stiffness, the damped frequency falls

Changing physical hardware can change more than one parameter. Record the complete configuration.

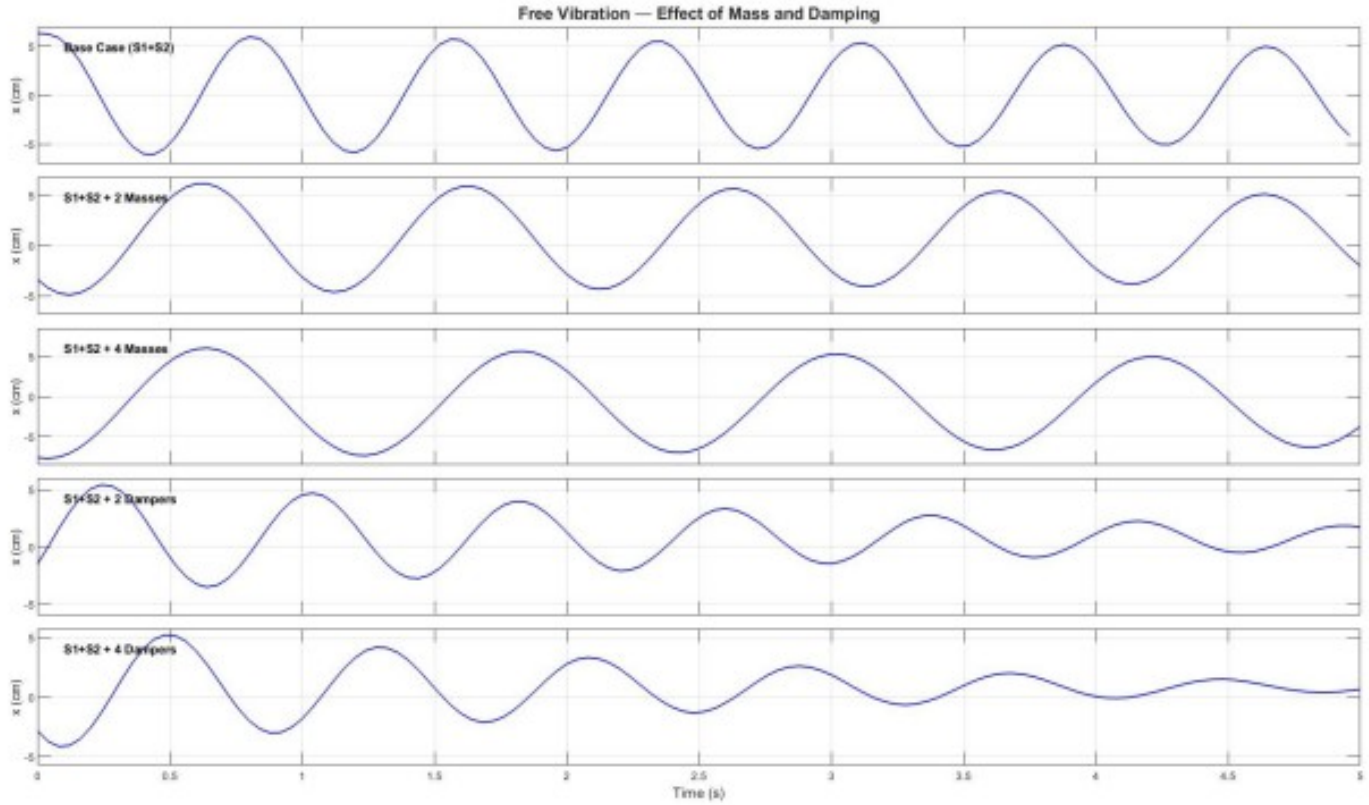
Uncertainty follows the measurements, not just the fit

Timing	Use multiple complete cycles; quantify time resolution and peak-selection variation.
Amplitude	Subtract equilibrium offset; choose same-sign peaks well above the noise floor.
Hardware	Include magnets, fasteners and attachments in moving mass; calibrate each spring.
Reporting	Compare predicted and measured values with units, residual plots and defensible significant figures.

A systematic mismatch is evidence about the apparatus or model, not simply “bad data.”

30–40 min

Prior-class measurements: mass and damping changes



Read the axes first

Horizontal: time [s]

Vertical: position [cm]

Upper rows: mass changes

Lower rows: brake changes

Every blue trace is displacement versus time. These are anonymized prior-class measurements.

Reading the historical plots: what each curve means

Blue x(t)	Recorded position versus time: spacing of same-sign peaks gives T_d; the shrinking heights give α.
Red spectrum	FFT magnitude of the same recorded run: horizontal location checks dominant frequency.
Not a damping test	FFT peak HEIGHT also changes with displacement amplitude, window length and normalization.
Provenance	These are anonymized prior-class experimental plots, not curves generated by the ideal model.

Always read axes and compare like-for-like configurations before inferring a trend.

41–42 min

The blue traces and red spectra answer different questions

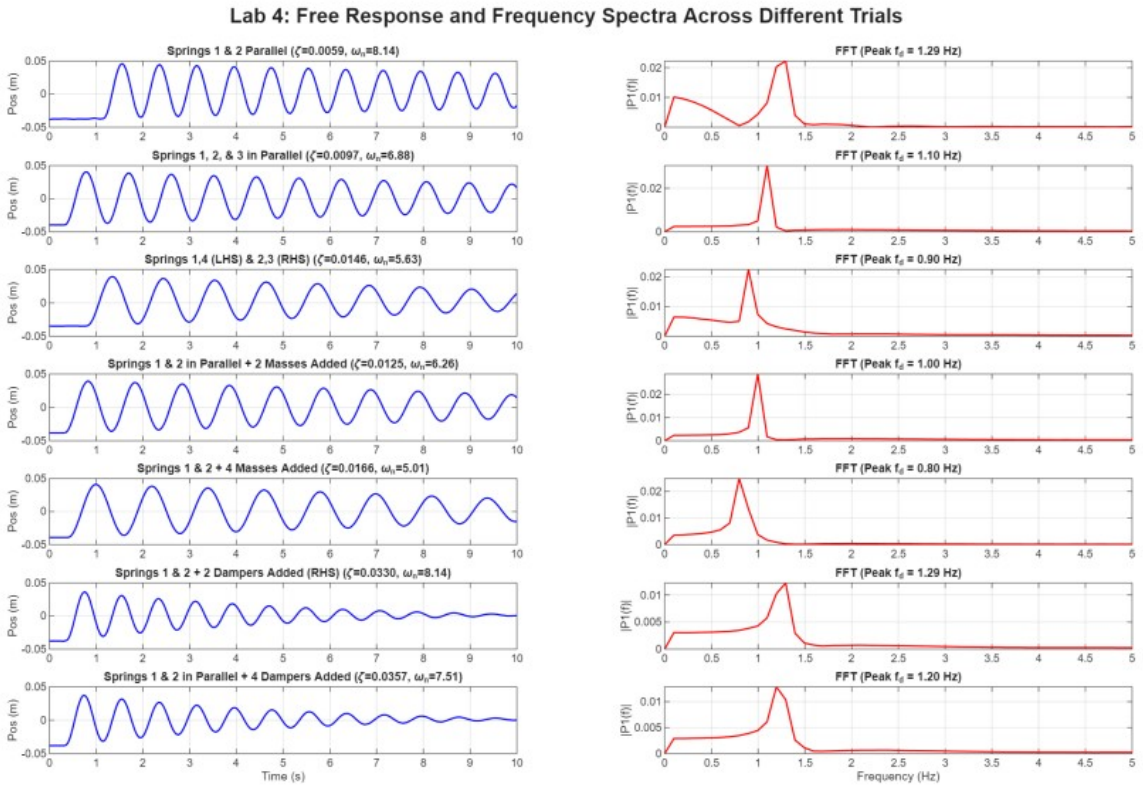


Figure 1: Lab 4 - Free Response and Frequency Spectra Across Different Trials

Blue: x(t)

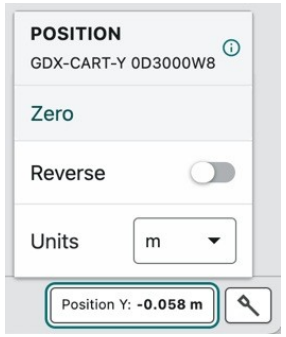
Peak spacing gives Td. Peak-height decay gives α.

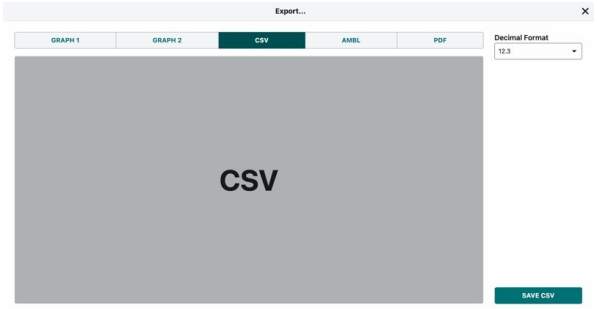
Red: FFT magnitude

Peak location checks fd. Peak height also depends on record and initial amplitude.

One row represents one configuration. Time trace and spectrum describe the same recorded motion.

Collection settings and the order of operations





1. Level the track and connect the correct serial number.

2. Center the cart and zero its position.

3. Use fs ≤ 25 Hz per the stated local lab procedure; record actual fs.

4. Start Collect, then release without a push.

5. Record several cycles and a measurable decay.

6. Export one CSV per case and plot it immediately.

Record total moving mass, spring IDs, brake count, sampling rate, initial displacement, and filename.

A reliable analysis begins with the raw record

1. Preserve the raw CSV. Select the free-response interval after release.
2. Subtract the equilibrium offset and inspect drift or clipping.
3. Find reliable equal-sign peaks above the noise floor.
4. Estimate Td and α, then calculate ωn, ζ, and c.
5. Compare the decay model with x(t), and check the FFT peak.

Run the main analysis script listed in the README. Helper functions require input arguments.

Uncertainty: what limits our parameter estimates?

Peak timing	For N cycles: Td = (t[i+N] − t[i])/N. Longer spans reduce the influence of individual timing errors.
Peak amplitudes	Offset errors, sensor noise, and late low-amplitude peaks bias the logarithmic decrement and log-peak slope.
Physical parameters	c = 2mα: uncertainty in moving mass propagates directly into c; spring calibration affects predicted ωn.
Report responsibly	Use repeat trials or fit-based uncertainty estimates; report units, assumptions, and meaningful precision.

Do not report more digits than the measurements justify.

A fitted curve must survive a residual check

Fit model	$X(t) = X_{eq} + \exp(-\alpha t)[C\cos(\omega_d t) + D\sin(\omega_d t)]$ after the release time.
Plot residuals	$r(t) = X_{measured}(t) - X_{fit}(t)$. Inspect for drift, periodic structure, asymmetry, and changing error.
Question assumptions	Dry friction, wheel slip, spring slack, or nonlinear magnetic braking may invalidate a constant-c model.
Cross-check	Compare peak-derived f_d with the FFT, and ω_n with $\sqrt{(k_L+k_R)/m}$ using measured hardware.

Agreement between independent checks is stronger evidence than a visually smooth fit.

Quality checks connect the mathematics to the hardware

Envelope does not decay smoothly	Check rubbing, slack springs, drift, and the noise floor.
FFT and peak timing disagree	Check units, record duration, FFT bins, and peak selection.
Theory and experiment disagree	Check total moving mass and keq = kL + kR.
Fitted curve misses the points	Fit a damped response with an offset, then inspect residuals.

Analyze one complete case and show the GTA before leaving.

What students should explain in Pre-Lab 4 and the report

Pre-Lab 4

Explain each model variable and its unit.

Predict changes in period and decay.

Run the provided notebook and interpret its plots.

Laboratory report

Explain how spring stiffness was calibrated.

Show x(t), fitted decay, and frequency spectra.

Compare measured and predicted parameters for all seven cases.

Use equations that support your analysis. Submit the pre-lab Word/PDF before your own lab section.

New • active learning

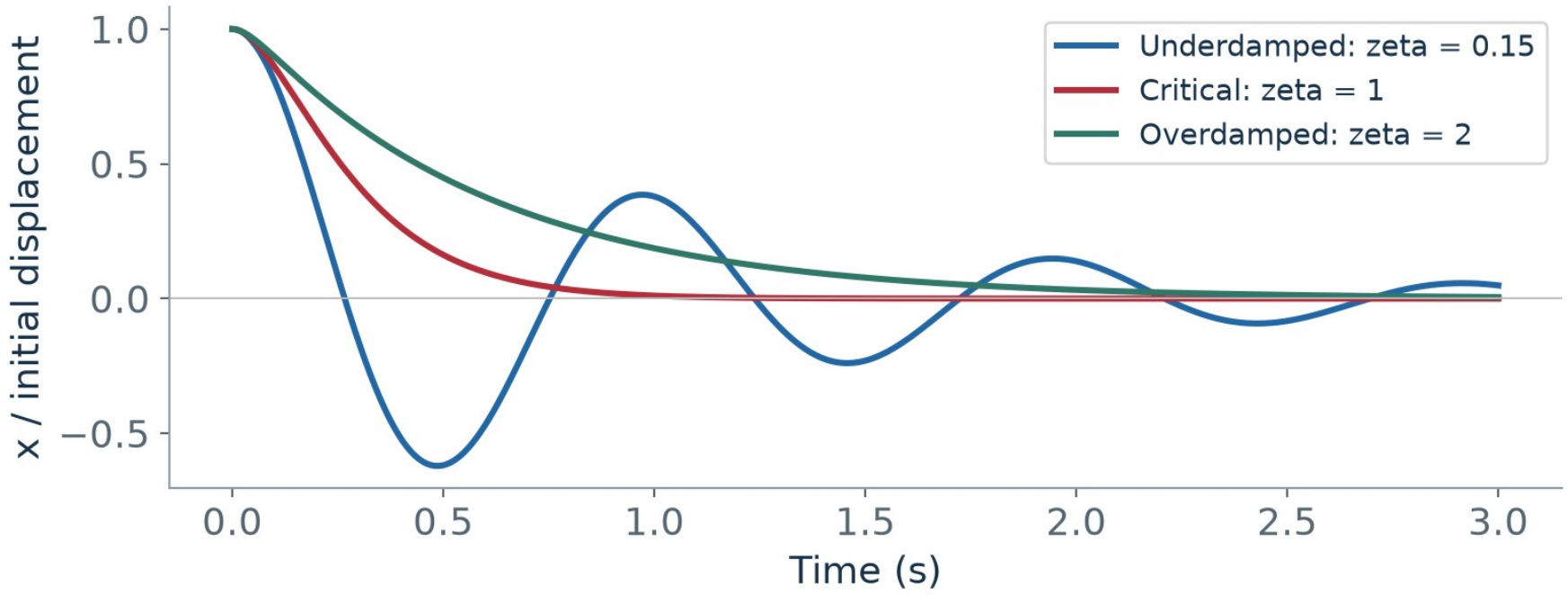
Exit ticket: predict, measure, defend

Prediction	If mass increases while k and c stay fixed, what happens to ωn and α?
Interpretation	If damping increases at fixed m and k, which frequency remains fixed: ωn or ωd?
Data decision	Which peaks would you exclude from a log-peak fit, and why?
Lab readiness	Can you identify the release time, Xeq, actual sampling rate, and all seven configurations?

Answers: ωn decreases; α decreases. ωn stays fixed; ωd decreases.

49–50 min

Appendix: three damping regimes from the same equation



The same m and keq produce different responses as c changes. Peak analysis requires underdamped motion.

Appendix: a worked peak-measurement example

$$N = 5, \quad t_{i+N} - t_i = 5.00 \text{ s}, \quad p_i = 0.040 \text{ m}, \quad p_{i+N} = 0.020 \text{ m}$$

$$T_d = 1.00 \text{ s}, \quad \omega_d = 6.283 \text{ rad/s}$$

$$\delta = \frac{\ln 2}{5} = 0.1386, \qquad \zeta = 0.0221$$

$$\omega_n = 6.285 \text{ rad/s}, \qquad c = 0.139 \text{ N s/m} \quad (m = 0.50 \text{ kg})$$

The observed and undamped frequencies are nearly equal here because damping is light.

Appendix: initial conditions determine the constants

$$x(0) = x_0, \qquad \dot{x}(0) = v_0, \qquad \alpha = \zeta\omega_n$$

$$C = x_0, \qquad D = \frac{v_0 + \alpha x_0}{\omega_d}$$

$$x(t) = e^{-\alpha t}\left[x_0\cos(\omega_d t) + \frac{v_0 + \alpha x_0}{\omega_d}\sin(\omega_d t)\right]$$

A release from rest means v0 = 0. It does not generally mean the sine coefficient is zero.

Appendix: exact spring correction and model limits

$$k = m_h \omega_{n,s}^2 = \frac{m_h \omega_{d,s}^2}{1 - \zeta_s^2}$$

Ignoring ζs is reasonable only when the hanging calibration motion has light damping.

A heavy spring adds distributed moving mass. Include a justified effective-mass correction if needed.

A spring going slack or rubbing invalidates the simple linear model.

The nominal spring constant is a reference. The laboratory measures each spring separately.

Sources and how the lecture uses them

Prior MIE 402 theory: Lecture Notes / 12_Lab4_Theory.pdf

Prior MIE 402 experiment: Lecture Notes / 13_Lab4_Exp.pdf

Prior laboratory procedure: LAB4_MIE402.pdf

Vernier Go Direct Sensor Cart manual: vernier.com/manuals/gdx-cart-g/

Vernier Eddy Current Brake: vernier.com/product/eddy-current-brake/

Prior-class figures are anonymized. Model plots in this lecture are computed examples.
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